

Hydrogen based energy storage and fuel cells

Researchers at the RIKEN Cluster for Pioneering Research in Japan have created a solid electrolyte that transports hydride ions (H^-) at room temperature, a breakthrough with implications for hydrogen-based solid-state batteries and fuel cells. This development promises improved safety, efficiency, and energy density, crucial for advancing a viable hydrogen-based energy economy. Hydrogen-based fuel cells use proton conduction through a polymer membrane, requiring constant hydration. To simplify designs, scientists aimed to conduct negative hydride ions through solid materials at room temperature. Kobayashi's team focused on lanthanum hydrides ($LaH_{3-\delta}$), known for their hydrogen properties. They overcame hydrogen non-stoichiometry by substituting lanthanum with strontium (Sr) and introducing oxygen, resulting in a formula ($La_{1-x}Sr_xH_{3-x-2y}O_y$) enabling efficient hydride ion conduction. Crystalline samples of the material showed high room-temperature hydride ion conduction.

Can a vibrating capsule help in controlling weight?

MIT engineers have developed an inventive ingestible capsule that induces stomach vibrations, replicating the sensation of fullness. In animal experiments, consuming this capsule 20 minutes prior to meals triggered the release of satiety hormones, resulting in a remarkable 40% reduction in food intake. The capsule, resembling a standard multivitamin in size, features an integrated vibrating component. Upon ingestion and exposure to gastric fluids, it activates mechanoreceptors, stimulating the vagus nerve and emulating post-meal hormone release patterns. The capsule vibrates for approximately 30 minutes, with the option of adjusting the duration wirelessly. Importantly, it safely passes through the digestive system, offering potential cost-effective and accessible benefits.

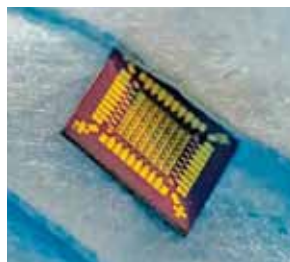


MIT engineers designed an ingestible capsule that vibrates within the stomach, creating an illusory sense of fullness and reducing appetite (Credit: Courtesy of the researchers, MIT News)

Silk-based transistors for hybrid applications

Researchers at Tufts University have found that integrating silk into transistors can create highly responsive and fast

sensors. They applied silk films of various thicknesses to chips and used silk as an insulator within the transistors.



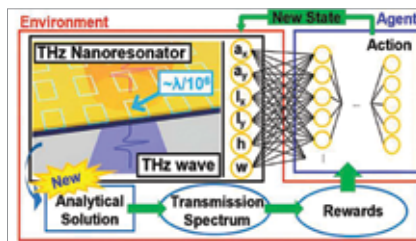
Hybrid biological transistors change their electronic behaviour in response to gases and other molecules in the environment (Credit: FIO OMENETTO/TUFTS University)

Transistors are composed of metal electrical leads connected to a semiconductor material with a gate acting as an electrical input, separated from the transistor's body by the insulator. When the gate reaches a specific voltage, it induces an electric field, allowing current flow. The nanoscale silk films respond to moisture in the air, altering electrical properties and making the transistors

function as compact, rapid humidity sensors. These hybrid devices have potential applications in detecting health-related conditions, including cardiovascular and pulmonary diseases, sleep apnea, and monitoring blood oxygen and sugar levels.

AI-powered terahertz breakthrough amplifies communication waves 30,000 times

A research team at UNIST has unveiled a technology capable of amplifying terahertz (THz) electromagnetic waves. This achievement, made possible through integrating artificial intelligence (AI) based on physical models, is poised to revolutionise 6G communication frequencies. The



Graphical abstract (Credit: Nano Letters, 2023)

team employed advanced optimisation techniques to tailor THz nano-resonators for 6G communication. These nano-resonators generated an electric field sur-

passing conventional electromagnetic waves by over 30,000 times, translating to a remarkable 300% increase in efficiency compared to previous THz nano-resonators. While AI-based inverse design was traditionally applied to visible or infrared optical device structures, adapting it to the 6G communication frequency range posed challenges due to its small scale. The team combined a THz nano-resonator with an AI-based inverse design method, enabling swift device optimisation on personal computers. This research carries promising implications for detectors, molecular detection sensors, and bolometer studies.